



* Practical - 16 *

* activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etUserId"
        android:hint="User ID"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp" />

    <Button
        android:id="@+id/btnLogin"
        android:text="Login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp" />

    <Button
        android:id="@+id/btnRegister"
        android:text="Register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp" />

    <TextView
        android:id="@+id/tvResult"
        android:text=""
        android:textSize="18sp"
        android:textColor="@android:color/holo_blue_dark"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

</LinearLayout>
```




* DBHelper.java

```
package com.example.authapp;
```

```
import android.content.ContentValues;
```

```
import android.content.Context;
```

```
import android.database.Cursor;
```

```
import android.database.sqlite.SQLiteDatabase;
```

```
import android.database.sqlite.SQLiteOpenHelper;
```

```
public class DBHelper extends SQLiteOpenHelper {
```

```
    private static final String DATABASE_NAME = "AuthDB";
```

```
    private static final int DATABASE_VERSION = 1;
```

```
    private static final String TABLE_USERS = "users";
```

```
    public DBHelper (Context context) {
```

```
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
```

```
    }
```

③ Override

```
    public void onCreate (SQLiteDatabase db) {
```

```
        String createTable = "CREATE TABLE " +
```

```
            TABLE_USERS + " (" +
```

```
                "id INTEGER PRIMARY KEY AUTOINCREMENT," +
```

```
                "userid TEXT UNIQUE," +
```

```
                "password TEXT)";
```

```
        db.execSQL(createTable);
```



```
ContentValues values = new ContentValues();  
values.put("userid", "admin");  
values.put("password", "1234");  
db.insert(TABLE_USERS, null, values);  
}
```

⑨ Override

```
public void onUpgrade(SQLiteDatabase db,  
                      int oldVersion, int newVersion) {  
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);  
    onCreate(db);  
}
```

```
public boolean insertUser(String userid, String password)  
{  
    SQLiteDatabase db = this.getWritableDatabase();  
    ContentValues values = new ContentValues();  
    values.put("userid", userid);  
    values.put("password", password);  
    long result = db.insert(TABLE_USERS,  
                           null, values);  
    return result != -1;  
}
```

```
public boolean checkUser(String userid, String password)  
{  
    SQLiteDatabase db = this.getReadableDatabase();  
    Cursor cursor = db.rawQuery("SELECT * FROM " +  
                                TABLE_USERS + " WHERE userid=?  
                                AND password=?",  
                                new String[] {userid, password});  
    boolean exists = cursor.getCount() > 0;  
    cursor.close();  
    return exists;  
}
```




* MainActivity.java

```
package com.example.authapp;
```

```
import androidx.appcompat.app.AppCompatActivity;  
import android.os.Bundle;  
import android.widget.*;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    EditText etUserId, etPassword;  
    Button btnLogin, btnRegister;  
    TextView tvResult;  
    DBHelper dbHelper;
```

⑨ Override

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);
```

```
    etUserId = findViewById(R.id.etUserId);  
    etPassword = findViewById(R.id.etPassword);  
    btnLogin = findViewById(R.id.btnLogin);  
    btnRegister = findViewById(R.id.btnRegister);  
    tvResult = findViewById(R.id.tvResult);
```

```
    dbHelper = new DBHelper(this);
```



```
btnLogin.setOnClickListener (v-> {  
    String userid = etUserId.getText().toString();  
    String password = etPassword.getText().toString();  
  
    if (dbHelper.checkUser(userid, password)) {  
        tvResult.setText(" ☒ Login Successful ! Welcome "  
            + userid );  
    } else {  
        tvResult.setText(" X Invalid Credentials ");  
    }  
});  
  
btnRegister.setOnClickListener (v-> {  
    String userid = etUserId.getText().toString();  
    String password = etPassword.getText().toString();  
  
    boolean inserted = dbHelper.insertUser(userid,  
                                                password);  
    if (inserted) {  
        tvResult.setText(" ☒ User Registered  
            Successfully ! ");  
    } else {  
        tvResult.setText(" X Registration Failed  
            (UserId already exists) ");  
    }  
});  
}
```